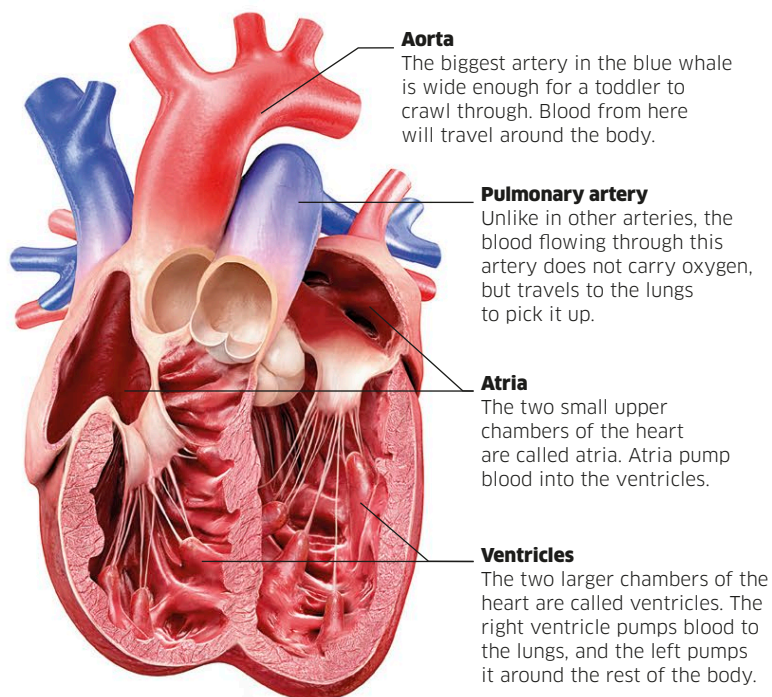


The heart

The blue whale has the biggest heart of any animal: weighing in at 400 lb (180 kg) and standing as tall as a 12-year-old child. Containing four chambers, it is made of solid muscle, and contracts with a regular rhythm to pump blood out through the body's arteries. When its muscles relax, the pressure inside the chambers dips very low to pull in blood from the veins.



Network of vessels

Thousands of miles of blood vessels run through the body of a blue whale. Thick-walled arteries (shown in red) carry blood away from the heart and thin-walled veins (shown in blue) ferry it back. These both branch off countless times to form a network of microscopic capillaries (smaller blood vessels) that run between the cells.

Arteries leading to the head provide oxygenated blood for the brain.

Circulation

Blood is an animal's essential life support system, transporting food and oxygen around its body and removing waste from cells.

Animals have trillions of cells that need support, and a vast network of tiny tubes called blood vessels stretches throughout their bodies in order to reach them all. A pumping heart keeps blood continually flowing through the blood vessels, and this bloodstream gathers food from the digestive system and oxygen from lungs or gills. When the blood reaches cells, these essentials pass inside, while waste moves back out of the cells and is then carried away by the blood to excretory organs, such as the kidneys.

11 tons of blood are contained within the body of a blue whale. Its heart pumps the equivalent of two baths with every beat.

Arteries

Arteries carry bright red blood full of oxygen away from the heart. The blood moves at high pressure, because it is propelled by the heart's strong beat.

Veins

Veins carry purplish-red blood back to the heart. It is harder for the blood to travel in this direction, so muscles push on the veins to help the blood move along.

